RITAM MUKHERJEE

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EDUCATION

University of Waterloo — Expected May 2030

Bachelor of Applied Science in Mechatronics Engineering (GPA: 3.8/4.0) — *Waterloo, Ontario*

- **Relevant Coursework:** Data Structures and Algorithms (C++), MTE 100 (Solidworks, AutoCAD, GD&T), Linear Algebra

EXPERIENCE

SuraNeoAge Systems Transformations (Early Stage Startup) — May 2026 – Aug 2026

Software Engineering Intern — *Remote*

- Fine-tuned Phi-4 Mini (14B params) via QLoRA for financial entity and topic extraction, using GPT-4o-generated ground-truth labels across 2,700+ scraped financial news articles enriched with National Stock Exchange sector data; outperformed the base model and GPT-4o mini in a 100-article blind ABC comparison judged by the team's business analyst
- Built the ingestion-to-training data pipeline (PostgreSQL, FastAPI, Docker) and fine-tuned FinBERT (110M params) for company NER, reaching 0.76 token F1 / 0.88 recall
- Built an on-demand LLM relevance-scoring endpoint replacing manual editorial curation; persisted latest verdict per article in Redis with automatic cache invalidation on reclassification, decoupling read latency from LLM inference time

University of Waterloo — Sep 2025 – May 2026

Undergraduate Student Researcher — *Waterloo, Ontario*

- Worked under Dr. Shirley Tang on early-stage development of a 7-axis 3D printer using a Gen3 robot arm — implemented ROS 2 Cartesian trajectory-tracking and homing scripts on the physical arm, while researching non-planar slicing and g-code conversion

University of Waterloo — Aug 2024 – Jun 2025

Volunteer Student Researcher — *Waterloo, Ontario*

- Assisted Dr. Eugene Li on an industrial CNC tool remaining-useful-life project — built a sensor-to-model data pipeline, prototyped a PyTorch Transformer, and trained an NCA+KNN classifier reaching 79.6% test accuracy on 1.5M+ accelerometer samples

KW Sandbox — Jul 2023 – Jun 2025

President — *Waterloo, Ontario*

- Led a 12-person team running technical STEM workshops for 160+ students, coordinating venues, sponsors, and speakers

PROJECTS

Personal Website | *https://www.ritammukherjee.ca/ — Next.js, React, TypeScript, Supabase, Vercel*

- Developed a full-stack portfolio featuring live integrations with Spotify and Discord APIs to display real-time user activity
- Implemented a custom messaging system using Supabase for the backend and integrated GitHub data to showcase dynamic project updates and repository statistics.

Axiom | *https://github.com/forkiron/axiom — Node.js, TypeScript, Supabase, Vercel*

- Built an AI platform to standardize Canadian school grades via test-difficulty analysis, a 3D interactive map, and a natural-language agent
- Developed full-stack with Next.js, Gemini AI, Mapbox, and Supabase with crowd-sourced adjustment data persistence

Motion Activated Door Monitoring System | *https://github.com/ryanli222/smartdoorbell — Raspberry Pi, OpenCV, Docker, Ngrok, FastAPI*

- Developed a real-time motion detection system using OpenCV on a Raspberry Pi to monitor a webcam feed and detect door activity.
- On detection, activated a display, streamed live video, played alert audio, enabled two-way audio, and uploaded images to FastAPI via Ngrok

1000 Elo Chessbot | *https://github.com/j3rry1iu/ChessHacks-Training — Python, CNNs, Numpy, HuggingFace*

- Built a chess engine using a ResNet style CNN, policy/value head, and a depth3 search to generate moves within 250ms.
- Used Modal cloud computing to train the model, uploading model weights and storing the model on HuggingFace.

TECHNICAL SKILLS

Languages: Python, C++, C#, TypeScript, JavaScript, SQL
ML/Data: PyTorch, HuggingFace, QLoRA, scikit-learn, Numpy, OpenCV, TensorFlow
Robotics: ROS 2, SolidWorks, AutoCAD, Fusion 360
Web/Infra: FastAPI, Next.js, React, PostgreSQL, Redis, Supabase, Docker, Vercel